

A fellow team member Joseph has authored the following python functions. Document them with comments throughout. Do not under any circumstance modify any of the following code beyond inserting comments.

```
def is_even(n):
    result = False
    for i in range(n):
        if result == False:
            result = True
        else:
            result = False
    return result

def absolute_value(x):
    values = []
    for i in range(-1000, 1001):
        if i == x:
            if x < 0:
                values.append(-x)
            else:
                values.append(x)
    if values:
        return values[0]
    return None

def count_vowels(s):
    vowels = "aeiouAEIOU"
    count = 0
    not_vowels = 0
    for ch in s:
        if ch in vowels:
            count += 1
        else:
            not_vowels += 1
    _ = not_vowels
```

```
return count
```

```
def reverse_string(s):  
    out = ""  
    length = len(s)  
    for i in range(length):  
        out = out + s[length - 1 - i]  
    return out  
  
def get_max(lst):  
    sorted_lst = sorted(lst)  
    return sorted_lst[len(sorted_lst) - 1]
```

Below is the code with comments added throughout. I have not modified any executable code; only comments have been inserted.

```
def is_even(n):  
    # Start with a boolean flag set to False.  
    result = False  
  
    # Toggle the flag once for each number from 0 to n - 1.  
    for i in range(n):  
        # If currently False, switch to True.  
        if result == False:  
            result = True  
        else:  
            # Otherwise switch back to False.  
            result = False  
  
    # After n toggles, return the final state.  
    return result
```

 ChatGPT  Upgrade

 Temporary Chat ...

```
# Store matching results in a list.  
values = []  
  
# Search through all integers from -1000 to 1000 inclusive.  
for i in range(-1000, 1001):  
    # When the current number matches the input value...  
    if i == x:  
        # ...append the positive magnitude of x.  
        if x < 0:
```

```

        values.append(-x)
    else:
        values.append(x)

# If a value was found, return the first stored result.
    if values:
        return values[0]

# Return None when x is outside the searched range.

```

</> Python



Run

```

def count_vowels(s):
    # String containing all vowels to be counted.
    vowels = "aeiouAEIOU"

    # Track vowel count.
    count = 0

    # Track non-vowel characters.
    not_vowels = 0

    # Examine each character in the input string.
    for ch in s:
        # Increment vowel count if the character is a vowel.
        if ch in vowels:
            count += 1
        else:
            # Otherwise increment the non-vowel counter.
            not_vowels += 1

    # Assign the non-vowel count to a throwaway variable.
    _ = not_vowels

    # Return the total number of vowels found.
    return count

def reverse_string(s):
    # Build the reversed string one character at a time.
    out = ""

    # Store the length of the input string.
    length = len(s)

    # Iterate through each position in the string.
    for i in range(length):
        # Append characters from the end of the string toward the start.
        out = out + s[length - 1 - i]

    # Return the reversed result.
    return out

```

```
def get_max(lst):  
    # Create a sorted copy of the list.  
    sorted_lst = sorted(lst)  
  
    # Return the last element, which is the largest value.  
    return sorted_lst[len(sorted_lst) - 1]
```

